1 Membrane: capacitor + batteries

Piece	Concept
Capacitor	Lipid between two salty baths. Q=CV. I_C=C dV/dt only while V changes. Zero at rest and AP peak. C=Aε/d ≈ 1 μF/cm². Bigger area → more C. Myelin ↑d → ↓C.
When is I_C=0?	Whenever dV/dt=0 (flat V). When is one ion's I_R=0? g=0 or V=E. Total I_R=0 at Vrest (currents cancel, channels are not all shut).
Battery	Concentration gradients from Na/K ATPase (3 Na out / 2 K in). Each ion's battery voltage is E_{ion} (Nernst). Spike ~10^6 ions vs ~10^{14} in cell → E's barely move. Pump refills over minutes. Ouabain ≠ instant silence.
E vs Vrest	Each ion's Nernst is nearly constant on spike timescales. Vrest is not constant: it is wherever I_{tot}=0. E_ion stays put; Vrest can slide.
Need both	Only C: everything drains to 0 mV. Only batteries: V jumps with no time course. C is fast/cheap. Batteries set where ions want to go.

2 Conductance, driving force, current

• g = 1/R ≈ (# open channels)×(one-channel g). Ease of flow right now. I–V slope is g. Parallel pores: g adds, R falls as 1/n. Channels = resistors; bilayer = capacitor.
• g is not DF. Current needs both: I = g(V−E). Closed: I=0 even if DF is huge. At E: I=0 even if every channel is open. Nernst = zero net current, not max current.
• DF = V−E pulls V toward that ion's E, not toward 0. K toward −90, Na toward +60. On an I–V, |DF| = distance from the x-intercept.
• g depends on voltage and time (gates). That is why voltage clamp holds V: so you watch time alone. Macro I = N · i · P_{open}. Channels flicker (stochastic); not a hard on/off switch.
• Voltage-gated current often exists only in a voltage window (P_open high in a range, then inactivation). Ligand-gated: transmitter, not S4.

Sign of current (+I = net positive charge leaving)

Ion / flux	Current sign
Na^+ or K^+ in / out	Inward = negative I. Outward = positive I.
Cl^- in / out	Cl^- entering (inward anion) = positive I. Cl^- leaving = negative I. Opposite of cations.
At Vrest	Inward I_{Na} and outward I_K are large and cancel. Rest is not "everything off."
At E_{Na}	I_{Na}=0 even if g_{Na} is huge. I_K can still be large (V far from E_K).
I–V two ions	Steeper slope owns Vrest. Combined slope = sum of g's (if g_{Na}=g_K, slope is 2g, I=0 midway).
Dashed "all open" line	Dashed = I if every channel open. Gap from a data point to that line = fraction actually open.

3 Nernst, GHK, rest

• Nernst: one ion at equilibrium. GHK / weighted g: several ions, none at eq, I_{tot}=0. P is not g (g also depends on how many are open and on V).
• Vrest sits next to the biggest g (usually K leak + a little Na → near E_K but positive to it). Extra g_{Na} → rest more positive. Raise $[K]_o$ → E_K and rest depolarize → more excitability, then Na inactivation / block.
• z=+1 K/Na, +2 Ca/Mg, −1 Cl/HCO_3. Same 10-fold: z=2 halves |E|. Anion: minus from z. E_{HCO3}≈−6 (slide +6 was wrong). Biology lives between E_{Na} and E_K. Undershoot ⇒ rest is not E_K. Overshoot past 0 ⇒ active Na, not "cell dying."

4 Channels and HH

Part	Job
S1–S4	Voltage-sensing domain. S4 = sensor (positive residues). Depolarization drives S4 out.
S5–S6 + P-loop	Pore. P-loop = selectivity filter (which ion), not the sensor.
K_v	4 separate 6-TM subunits (tetramer), 24 TM, g_K=n^4. Ball-and-chain is on one N-terminus of A-type K, not delayed rectifier. Stochastic.
Na_v / Ca_v	1 polypeptide, 4 domains, 24 TM. g_{Na}=m^3h. Same architecture, fused. α2δ/gabapentin: Ca trafficking, not the pore.
m^3h picture	3 activation gates (m) + 1 IFM plug (h). Ions flow only if all 3 m's are on and IFM is not in the pore. h starts available (~1) at rest.
n^4 kinetics	Need all 4 on → sigmoid delay on open. Any 1 off closes it → ~4× faster close.
Act / deact / inact	Activation = m or n up. Deactivation = they go back down on repolarization. Inactivation = h or ball plugging an open channel. Not synonyms.
Evidence for ball	Internal trypsin (not bath); N-terminal deletion; peptide rescue. Killing inactivation → extra/broader spikes.

5 Clamp, AP, threshold

• V-clamp holds V, measures I. After a step I_C is a dying blip; leftover I is ionic. Early inward = Na (reverses ~E_{Na}, TTX). Late outward = K. Tail: I jumps with DF, then gates close. Instantaneous I–V slope = g now.
• I-clamp injects I, measures V (rest, AP, subthreshold). Free AP: I_{tot}≈0; ionic I is charging C.
• Peak ≠ peak I_{Na} ≠ peak g. At the peak dV/dt=0 so I_C=0 so net ionic I=0: $|I_{Na}|$≈$|I_K|$. I=g×DF: peak I_K is often before peak g_K (V collapsing toward E_K). Near overshoot Na DF is small; on the way down DF grows so I_{Na} can rise while g falls until h kills it.

Three stimulus strengths

Stimulus	What you see
Strong	Early, large regenerative AP.
Weak but supra	AP later and often slightly smaller. Threshold is a race, not −30 mV.
Subthreshold	Graded depolarization that decays. Not an AP unless it propagates to the terminal.

• College threshold: inward I_{Na} > outward I_K, then it must travel. Moves with h, leftover g_K, dV/dt (slow ramp lets K catch up), $[K]_o$, T, AIS Na density. AIS/hillock starts spikes because of high Na_v density, not "most depolarized." Refractory: absolute (h≈0) vs relative (extra g_K).

6 Why an AP · cable · myelin

• Dendrites already compute in analog; that wave decays (r_a, leak, C). The AP regenerates from local batteries so the terminal still gets a spike. Not hydraulics: faster, cheaper (ions not bulk fluid), branches, can invert sign chemically.
• Leading-edge Na in charges the next patch's C. Most charge is used locally; leftover down the axoplasm depolarizes ahead. Trailing: K out + Na inactivated.
• Naive axon, mid-stim → both ways. After passage → no bounce-back (h low). Soma is a huge C so backward wave often dies. Dendritic Na → back-propagating APs (NMDA/Mg is 101B). Branch = extra area → possible failure.
• Slower: thin (high r_a), leaky, high C, cold, few/inactivated Na_v, nodes too far. Faster: fat axon (r_a falls as r^2; extra C only as r — squid; human proprioception not pain), myelin, warm.
• Myelin: CNS oligodendrocyte, PNS Schwann. ↑R_m ↓C so less leak and fewer charges per mV (Q=CV). Internode = passive and fast. Regeneration only at nodes (clustered Na_v). Whole internode still depolarizes — not a magic hop. Demyelinate: C returns AND no internodal Na_v → block.
• τ ≈ $R_m C_m$ (charging time). λ ~ √(R_m/r_a) (passive distance). Open channels lower R_m.

7 Synapse, vesicles, last slide

• Golgi (continuity, like vessels) vs Cajal (separate cells): both a bit right. Gap jx share cytoplasm/ions/small molecules (cyt c). Neighbor follows your ΔV. Cannot invert sign. Chemical can (Glu vs GABA).
• AP → Ca_v at active zone → Ca in (mM out / ~100 nM in) → vesicle fusion (~30–50 nm) → transmitter → receptors. Ionotropic = ligand-gated pore (fast). Metabotropic/GPCR = no pore. Vertebrate NMJ = ACh. Workhorses: Glu excite, GABA/glycine inhibit.
• Why vesicles, not "as made": integer quantum, timed to the AP (Ca), placed at the cleft, recyclable, p≈0.2–0.5 per CNS zone so many weak votes. NMJ uses thousands of zones. Katz: minis = 1 vesicle; low-Ca evoked = 0,1,2,3×.
• Last slide: thousands of tiny inputs; AIS asks fire? If you only count time, any 3 of 4 coinciding can spike. Nearby synapses share cable — two coincident and close can beat two far ones. Output = when and where. Most of the trace is subthreshold on purpose: that analog mess is the computation.
• ChR2 = light-gated cation channel (not a pump). Boyden Fig. 2: millisecond spike timing / low jitter, not just "cells fire in the light."

8 How a stem maps onto the above

If you see…	Think…
I vs V	Slope=g. Intercept=E or Vrest. Side shift=E changed. Rotate=g changed. Add slopes if two ions.
V vs t	AP phases. Overshoot=active Na. Undershoot ⇒ rest≠E_K. Aborted = threshold race. Strong/weak/subthreshold as in §5.
I vs t after a V step	I_C blip dies. Early Na, late K. Reverse of early I ≈ E_{Na}. Tail I jumps with DF.
g, n, m, h vs t or dashed line	n/m start low, h starts high. Peak I≠peak g. Dots vs dashed = fraction open.
V vs distance	Passive decays. Regenerative keeps shape. Dies ⇒ not an AP.
Concentrations / z	Nernst. Divide by z. Anion minus. Don't use Nernst for a real cell's Vrest.
mV and nA	Ohm. Convert to V and A first. 100 mV / 1 nA = 100 MΩ.
Area, radius, "how many ions"	C=C_{spec}A, Q=CV, n=Q/(z e). I_C only while V changes.
"Change $[K]_o$ / $\bar{g}$ / TTX / trypsin / myelin"	Toward that ion's E; TTX no rise; TEA broader; no inactivation → extra spikes; lose myelin = C back + no Na_v.
"Properties of / voltage-dependent because / sensor"	Resistors. P(open) vs V, not I_open. Sensor=S4, filter=P-loop, ball=N-term/IFM. Na=1 protein, K=tetramer.
"Current is zero"	Which: I_C, one ion, or I_{tot}?
Why / evidence / last slide	Three causal steps. Ball: trypsin, deletion, peptide. Vesicles vs drip. Integration = when and where. Why AP: analog cable decays.

9 Formulas

Ohm V = IR = I/g g=1/R

Means	V difference; I charges/s; R how hard; g how easy.
When	Numbers for V,I; I–V slope; parallel channels.
Solve	Convert to V and A. R=V/I. n copies: g_{tot}=n g, I=n I.

Current I = g (V − E)

Means	DF=(V−E). +I = +charge out. Cation: V>E → out.
When	Direction; I at a V; why I≠g; AP peak; clamp.
Solve	Get E, get V. g=0 or V=E → I=0. Sign flips for anions.

Nernst E=(58–62 mV)/z · $\log_{10}$(o/i)

Means	ln: (RT/zF)ln(o/i). 37 °C≈61.5. z=+1,+2,−1.
When	One ion's reversal. Magne/Kali. $[K]_o$ change.
Solve	out/in → z → $\log_{10}$ → ×(62/z). 10× ⇒ |log|=1. Not for Vrest of a real cell.

Rest Vrest = Σ (g_i/g_{tot}) E_i (I_{tot}=0)

Means	GHK: same tug-of-war with P's.
When	Several ions; more Na open; why −70 not −90.
Solve	Biggest g wins. Equal g_{Na}=g_K → midway, slope 2g.

Capacitor Q=CV I_C=C dV/dt C=Aε/d I_{tot}=I_R+I_C

Means	I_C only while V changes. C_{spec}≈1 μF/cm².
When	How many ions; myelin; AP peak.
Solve	Sphere A=4πr² (r in cm). Q=CV (V in volts). n=Q/(z e). Peak: dV/dt=0 ⇒ I_{Na}=−I_K.

HH g_K=$n^4\bar{g}_K$ g_{Na}=m^3h $\bar{g}_{Na}$ then I=g(V−E)

Means	n,m,h∈[0,1]. n^4 delay on, fast off. m^3h = 3 act + 1 inact.
When	Transient Na; traces; trypsin (h stuck).
Solve	Depolarize: m↑ fast, h↓ slower, n↑ slow. Open Na = high m AND still-high h. Then I still follows DF.

Cable τ≈R_mC_m λ~√(R_m/r_a) I_{macro}=N·i·P_{open}

Means	τ charging time. λ passive distance. i one-channel I.
When	Decay, myelin, diameter, flicker vs smooth I.
Solve	Which of C, R_m, r_a moved? Next patch still needs I_{Na}>I_K.

10 Constants

Convert first. mV÷1000=V. nA=10^{-9} A. pA=10^{-12}. MΩ=10^6 Ω. nS=10^{-9} S. pF=10^{-12} F. 1 nA×1 ms=1 pC=6.25×10^6 charges (÷z for ions). e=1.6×10^{-19} C. R=8.31. F=96485. 20 °C→~58 mV; 37 °C→~61–62. 10×: z=1 → |E|≈58–62 mV. z=2 → ≈29–31. 100× z=1 → ≈116–124. $\log_{10}10$=1, $\log_{10}0.1$=−1.
E (mV): E_K≈−90. E_{Cl}≈−70. Vrest≈−70. E_{HCO3}≈−6. E_{Na}≈+55 to +66. E_{Ca}≈+120 to +135. 10× Mg in>out z=2 ≈−31. K 5/100 at 37 °C ≈−80. Conc: K 140/5. Na 15/145. Cl 8/110. Ca 100 nM / 1–2 mM. 10 μm sphere → C≈12.6 pF; 60 mV → ~4.7×10^6 ions. p≈0.2–0.5 vesicle/AP/CNS zone. Applet: nerve.bsd.uchicago.edu

11 Practice short answers (Q1–9)

Q	Write this
1 Na vs K	Same: 24 TM (S1–S6 × 4), S4 sensor (positive residues), S5–S6 + P-loop pore. Depolarization drives S4 out → P(open)↑. Diff: K_v = tetramer of four 6-TM proteins, g_K=n^4. Na_v = one polypeptide, 4 domains, g_{Na}=m^3h + fast IFM. Sensor = S4, not P-loop (filter) and not the ball (inact. plug). Both have four S4s — Na is not "3 S4s open / 1 S4 close." That 3+1 is m^3h.
2 K structure	Tetramer: 4 separate 6-TM subunits, 24 TM, one shared pore. S1–S4 = voltage-sensing domain; S4 = sensor. S5–S6 + P-loop = pore. Mouth lined by the four P-loops (carbonyls coordinate dehydrated K^+). n^4: all 4 activation particles on to conduct; any 1 off shuts it. A-type K has an N-terminal ball; delayed rectifier mostly does not on spike timescales.
3 Ball evidence	(1) Internal trypsin (not bath) clips the N-ball → inactivation lost, current stays on, extra/broader spikes. (2) N-terminal deletion kills inactivation; a ball peptide rescues it (plug need not stay covalently attached). Na_v: same idea with IFM in an intracellular loop, not a free N-ball on a separate subunit.
4 Subthreshold ΔV	Those wiggles are the computation. Minis, EPSPs/IPSPs, leftovers, channel noise — each too small to spike. Sum in time and space at the AIS (high Na_v) decides whether I_{Na}>I_K and whether a spike propagates. Last slide: when and where. If every little depolarization were an AP you would not need a dendritic tree.
5 Threshold	Not a fixed mV. College: inward I_{Na} > outward I_K, then it must travel. Same V can spike or abort. Moves with h (Na availability / refractory), leftover g_K, dV/dt (slow ramp lets K catch → higher threshold), $[K]_o$, T, AIS Na density, stochastic gating. Nothing holds it at −30.
6/8 Conc + pump	Concentrations = Nernst batteries (E_K≈−90, E_{Na}≈+60). Rest is the g-weighted average at I_{tot}=0 — near E_K but positive to it (~−70). More g wins. Extra g_{Na} → rest more positive. Pump (3 Na out / 2 K in) maintains gradients over minutes; it is not the AP motor. Ouabain ≠ instant silence. One spike ~10^6 ions vs ~10^{14} in the cell → E's barely move.
7 Peak dV/dt=0	I_{tot}=I_C+I_{ionic} and I_C=C dV/dt. Peak: dV/dt=0 ⇒ I_C=0 ⇒ net ionic I=0. Strong inward I_{Na} and strong outward I_K are equal and opposite. Channels are not off. Peak ≠ peak I_{Na} ≠ peak g. Then K wins and V falls.
9 $[K]_o$	↑$[K]_o$ → E_K less negative (Nernst log(o/i) toward 0) → rest (mostly K leak) depolarizes → closer to threshold → more excitable. Too much: h drops (Na inactivation) → cannot fire. ↓$[K]_o$ hyperpolarizes rest → less excitable. Nernst/rest effect, not the pump. Glia buffer K so one cell's efflux does not hyperexcite neighbors.